Assignment No.2

Implementation of DDL and DML Commands: **Name: Mahadev Suresh Jundale**

1.DDL Commands:

mysql> create database emp;

Query OK, 1 row affected (0.01 sec)

mysql> use emp;

Database changed

mysql> create table employee(id integer primary key,Emp_Name varchar(100),Dept varchar(100),Designation varchar(100),joiningDate date);

Query OK, 0 rows affected (0.05 sec)

```
mysql> desc employee;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| id          | int          | NO   | PRI | NULL    |       |
| Emp_Name    | varchar(100) | YES  |     | NULL    |       |
| Dept        | varchar(100) | YES  |     | NULL    |       |
| Designation | varchar(100) | YES  |     | NULL    |       |
| joiningDate | date         | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

mysql> insert into employee values(1,'Mahadev','IT','Manager','2025-01-01');

Query OK, 1 row affected (0.01 sec)

```
mysql> select * from employee;
+----+----------+------+-------------+-------------+
| id | Emp_Name | Dept | Designation | joiningDate |
+----+----------+------+-------------+-------------+
|  1 | Mahadev  | IT   | Manager     | 2025-01-01  |
+----+----------+------+-------------+-------------+
1 row in set (0.00 sec)
```

mysql> alter table employee add contact_No varchar(100);

Query OK, 0 rows affected (0.03 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> desc employee;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| id          | int          | NO   | PRI | NULL    |       |
| Emp_Name    | varchar(100) | YES  |     | NULL    |       |
| Dept        | varchar(100) | YES  |     | NULL    |       |
| Designation | varchar(100) | YES  |     | NULL    |       |
| joiningDate | date         | YES  |     | NULL    |       |
| contact_No  | varchar(100) | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

mysql> alter table employee drop contact_No;

Query OK, 0 rows affected (0.02 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> desc employee;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| id          | int          | NO   | PRI | NULL    |       |
| Emp_Name    | varchar(100) | YES  |     | NULL    |       |
| Dept        | varchar(100) | YES  |     | NULL    |       |
| Designation | varchar(100) | YES  |     | NULL    |       |
| joiningDate | date         | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

mysql> alter table employee modify Emp_Name varchar(200);

Query OK, 0 rows affected (0.03 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> desc employee;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| id          | int          | NO   | PRI | NULL    |       |
| Emp_Name    | varchar(200) | YES  |     | NULL    |       |
| Dept        | varchar(100) | YES  |     | NULL    |       |
| Designation | varchar(100) | YES  |     | NULL    |       |
| joiningDate | date         | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

mysql> alter table employee rename column Dept to dept_name;

Query OK, 0 rows affected (0.03 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> desc employee;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| id          | int          | NO   | PRI | NULL    |       |
| Emp_Name    | varchar(200) | YES  |     | NULL    |       |
| dept_name   | varchar(100) | YES  |     | NULL    |       |
| Designation | varchar(100) | YES  |     | NULL    |       |
| joiningDate | date         | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

mysql> rename table employee to emp;

Query OK, 0 rows affected (0.03 sec)

```
mysql> desc emp;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| id          | int          | NO   | PRI | NULL    |       |
| Emp_Name    | varchar(200) | YES  |     | NULL    |       |
| dept_name   | varchar(100) | YES  |     | NULL    |       |
| Designation | varchar(100) | YES  |     | NULL    |       |
| joiningDate | date         | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

mysql> truncate table emp;

Query OK, 0 rows affected (0.04 sec)

```
mysql> desc emp;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| id          | int          | NO   | PRI | NULL    |       |
| Emp_Name    | varchar(200) | YES  |     | NULL    |       |
| dept_name   | varchar(100) | YES  |     | NULL    |       |
| Designation | varchar(100) | YES  |     | NULL    |       |
| joiningDate | date         | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

mysql> select * from emp;

Empty set (0.00 sec)

mysql> drop table emp;

Query OK, 0 rows affected (0.03 sec)

mysql> create table employee(id integer primary key,Emp_Name varchar(100),Dept varchar(100),Designation varchar(100),joiningDate date);

Query OK, 0 rows affected (0.03 sec)

mysql> create table emp as select * from employee where 1=0;

Query OK, 0 rows affected (0.04 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> select * from emp;

Empty set (0.00 sec)

mysql> insert into employee values(1,'Mahadev','IT','Manager','2025-01-01');

Query OK, 1 row affected (0.01 sec)

```
mysql> select * from emp1;
+----+----------+------+-------------+-------------+
| id | Emp_Name | Dept | Designation | joiningDate |
+----+----------+------+-------------+-------------+
|  1 | Mahadev  | IT   | Manager     | 2025-01-01  |
+----+----------+------+-------------+-------------+
1 row in set (0.00 sec)
```

mysql> create table emp1 as select * from employee;

Query OK, 1 row affected (0.03 sec)

Records: 1 Duplicates: 0 Warnings: 0

```
mysql> select * from emp1;
+----+----------+------+-------------+-------------+
| id | Emp_Name | Dept | Designation | joiningDate |
+----+----------+------+-------------+-------------+
|  1 | Mahadev  | IT   | Manager     | 2025-01-01  |
+----+----------+------+-------------+-------------+
1 row in set (0.00 sec)
```

mysql> create table employee1(id integer primary key,Emp_Name varchar(100),Dept varchar(100),Designation varchar(100),joiningDate date)comment='Store department details';

Query OK, 0 rows affected (0.03 sec)

mysql> select * from employee1;

Empty set (0.00 sec)

2.DML Commands:

mysql> create database msj;

Query OK, 1 row affected (0.02 sec)

mysql> use msj;

Database changed

mysql> create table employee(id integer primary key,Name varchar(100),Department varchar(100),salary int);

Query OK, 0 rows affected (0.03 sec)

```
mysql> desc employee;
+------------+--------------+------+-----+---------+-------+
| Field      | Type         | Null | Key | Default | Extra |
+------------+--------------+------+-----+---------+-------+
| id         | int          | NO   | PRI | NULL    |       |
| Name       | varchar(100) | YES  |     | NULL    |       |
| Department | varchar(100) | YES  |     | NULL    |       |
| salary     | int          | YES  |     | NULL    |       |
+------------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

mysql> insert into employee values(1,'Mahadev','IT',90000);

Query OK, 1 row affected (0.01 sec)

mysql> select * from emloyee;

ERROR 1146 (42S02): Table 'msj.emloyee' doesn't exist

mysql> select * from employee;

```
+----+---------+------------+--------+
| id | Name    | Department | salary |
+----+---------+------------+--------+
|  1 | Mahadev | IT         |  90000 |
+----+---------+------------+--------+
1 row in set (0.00 sec)
```

mysql> insert into employee(id,Name,Department,salary) values(2,'Sahil','IT',80000);

Query OK, 1 row affected (0.01 sec)

```
mysql> select * from employee;
+----+---------+------------+--------+
| id | Name    | Department | salary |
+----+---------+------------+--------+
|  1 | Mahadev | IT         |  90000 |
|  2 | Sahil   | IT         |  80000 |
+----+---------+------------+--------+
2 rows in set (0.00 sec)
```

mysql> insert into employee values(3,'Vijay','IT',95000),(4,'Soham','IT',85000);

Query OK, 2 rows affected (0.01 sec)

Records: 2 Duplicates: 0 Warnings: 0

```
mysql> select * from employee;
+----+---------+------------+--------+
| id | Name    | Department | salary |
+----+---------+------------+--------+
|  1 | Mahadev | IT         |  90000 |
|  2 | Sahil   | IT         |  80000 |
|  3 | Vijay   | IT         |  95000 |
|  4 | Soham   | IT         |  85000 |
+----+---------+------------+--------+
4 rows in set (0.00 sec)
```

```
mysql> select * from employee where salary>85000;
+----+---------+------------+--------+
| id | Name    | Department | salary |
+----+---------+------------+--------+
|  1 | Mahadev | IT         |  90000 |
|  3 | Vijay   | IT         |  95000 |
+----+---------+------------+--------+
2 rows in set (0.00 sec)
```

```
mysql> select * from employee where salary>85000 and Department='IT';
+----+---------+------------+--------+
| id | Name    | Department | salary |
+----+---------+------------+--------+
|  1 | Mahadev | IT         |  90000 |
|  3 | Vijay   | IT         |  95000 |
+----+---------+------------+--------+
2 rows in set (0.00 sec)
```

```
mysql> select * from employee where Name like 'M%';
+----+---------+------------+--------+
| id | Name    | Department | salary |
+----+---------+------------+--------+
|  1 | Mahadev | IT         |  90000 |
+----+---------+------------+--------+
1 row in set (0.00 sec)

mysql> select * from employee where salary between 85000 AND 90000;
+----+---------+------------+--------+
| id | Name    | Department | salary |
+----+---------+------------+--------+
|  1 | Mahadev | IT         |  90000 |
|  4 | Soham   | IT         |  85000 |
+----+---------+------------+--------+
2 rows in set (0.00 sec)
```

```
mysql> select * from employee order by salary desc;
+----+---------+------------+--------+
| id | Name    | Department | salary |
+----+---------+------------+--------+
|  3 | Vijay   | IT         |  95000 |
|  1 | Mahadev | IT         |  90000 |
|  4 | Soham   | IT         |  85000 |
|  2 | Sahil   | IT         |  80000 |
+----+---------+------------+--------+
4 rows in set (0.00 sec)
```

```
mysql> select * from employee order by salary asc;
+----+---------+------------+--------+
| id | Name    | Department | salary |
+----+---------+------------+--------+
|  2 | Sahil   | IT         |  80000 |
|  4 | Soham   | IT         |  85000 |
|  1 | Mahadev | IT         |  90000 |
|  3 | Vijay   | IT         |  95000 |
+----+---------+------------+--------+
4 rows in set (0.00 sec)
```

mysql> update employee set salary=95000 where id=1;

Query OK, 1 row affected (0.01 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
mysql> select * from employee;
+----+---------+------------+--------+
| id | Name    | Department | salary |
+----+---------+------------+--------+
|  1 | Mahadev | IT         |  95000 |
|  2 | Sahil   | IT         |  80000 |
|  3 | Vijay   | IT         |  95000 |
|  4 | Soham   | IT         |  85000 |
+----+---------+------------+--------+
4 rows in set (0.00 sec)
```

mysql> update employee set salary=90000 where department='IT';

Query OK, 4 rows affected (0.01 sec)

Rows matched: 4 Changed: 4 Warnings: 0

```
mysql> select * from employee;
+----+---------+------------+--------+
| id | Name    | Department | salary |
+----+---------+------------+--------+
|  1 | Mahadev | IT         |  90000 |
|  2 | Sahil   | IT         |  90000 |
|  3 | Vijay   | IT         |  90000 |
|  4 | Soham   | IT         |  90000 |
+----+---------+------------+--------+
4 rows in set (0.00 sec)
```

mysql> update employee set salary=95000;

Query OK, 4 rows affected (0.01 sec)

Rows matched: 4 Changed: 4 Warnings: 0

```
mysql> select * from employee;
+----+---------+------------+--------+
| id | Name    | Department | salary |
+----+---------+------------+--------+
|  1 | Mahadev | IT         |  95000 |
|  2 | Sahil   | IT         |  95000 |
|  3 | Vijay   | IT         |  95000 |
|  4 | Soham   | IT         |  95000 |
+----+---------+------------+--------+
4 rows in set (0.00 sec)
```

mysql> delete from employee where id=1;

Query OK, 1 row affected (0.01 sec)

```
mysql> select * from employee;
+----+-------+-------------+--------+
| id | Name  | Department | salary |
+----+-------+-------------+--------+
|  2 | Sahil | IT          |  95000 |
|  3 | Vijay | IT          |  95000 |
|  4 | Soham | IT          |  95000 |
+----+-------+-------------+--------+
3 rows in set (0.00 sec)
```

mysql> create table employee(id integer primary key,Name varchar(100),Department varchar(100),salary int);

Query OK, 0 rows affected (0.03 sec)

mysql> insert into employee values(3,'Vijay','IT',95000),(4,'Soham','IT',85000);

Query OK, 2 rows affected (0.01 sec)

Records: 2 Duplicates: 0 Warnings: 0

```
mysql> select * from employee;
+----+-------+------------+--------+
| id | Name  | Department | salary |
+----+-------+------------+--------+
|  3 | Vijay | IT         |  95000 |
|  4 | Soham | IT         |  85000 |
+----+-------+------------+--------+
2 rows in set (0.00 sec)

mysql> select count(*) from employee;
+----------+
| count(*) |
+----------+
|        2 |
+----------+
1 row in set (0.01 sec)

mysql> select sum(salary) from employee;
+-------------+
| sum(salary) |
+-------------+
|      180000 |
+-------------+
1 row in set (0.00 sec)
```

```
mysql> select avg(salary) from employee;
+-------------+
| avg(salary) |
+-------------+
|  90000.0000 |
+-------------+
1 row in set (0.00 sec)
```

```
mysql> select max(salary),min(salary) from employee;
+-------------+-------------+
| max(salary) | min(salary) |
+-------------+-------------+
|       95000 |       85000 |
+-------------+-------------+
1 row in set (0.00 sec)

mysql> select * from employee limit 2;
+----+-------+------------+--------+
| id | Name  | Department | salary |
+----+-------+------------+--------+
|  3 | Vijay | IT         |  95000 |
|  4 | Soham | IT         |  85000 |
+----+-------+------------+--------+
2 rows in set (0.00 sec)
```